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The genetic edge

From injury prevention to talent identification, sports genomics is **reshaping** high-performance systems around the world.

“Genes do not determine champions, but they can help create them.”

In modern sport, victory is often decided by margins too small for the naked eye to detect. A hundredth of a second at the finish line, a fingertip deflection in the last minute, or a single lapse in concentration can alter the course of a match or a career. Behind these moments lie years of training, discipline, and increasingly, the influence of science. Sports science has become an indispensable part of elite competition, driving innovations in training, nutrition, recovery, and injury prevention. Among the most intriguing developments in this space is sports genomics, a field that studies how an athlete's genetic make-up influences their performance capabilities, adaptability to training, recovery patterns, and vulnerability to injury.

Sports genomics works by identifying genetic markers that affect traits such as muscle fibre composition, aerobic potential, bone density, inflammatory response, and nutrient metabolism. These insights are not guesses but the result of years of research linking specific genetic variations to physical attributes. For example, the ACTN3 gene is often associated with the presence of fast-twitch muscle fibres, giving certain athletes a natural advantage in explosive power sports. Other genetic indicators influence endurance capacity, injury susceptibility, and recovery speed. By decoding these factors, sports scientists can design training, nutrition, and recovery programmes that align with each athlete's biological profile.

What was once limited to academic studies has now reached the training grounds of the world's top sporting nations. The United States, the United Kingdom, Australia, and China have integrated sports genomics into their high-performance systems. In Australia, the Australian Institute of Sport has used genomic testing to optimise swimmer recovery strategies and fine-tune the endurance training of long-distance runners. UK Sport

has explored genetic profiling for cycling, rowing, and track athletes, using it to individualise strength programmes and prevent overtraining. In China, genomic analysis has been part of Olympic preparation in rowing and middle-distance running, helping athletes peak during major competitions while minimising injury risk. In the United States, professional leagues such as the NFL and NBA have worked with sports scientists to assess genetic predispositions to certain injuries, adapting player conditioning programmes to reduce time lost to injury.

In India, this technology is still largely untapped. The country's sporting achievements have grown in recent years, with notable success in cricket, badminton, wrestling, boxing, and athletics. Yet India continues to struggle with a high incidence of preventable injuries, inconsistent performance peaks, and a lack of structured systems for early talent identification. Many promising careers have been interrupted, or in some cases ended, by injuries that might have been avoided with better monitoring and targeted training. The adoption of sports genomics could change this dynamic by providing precise, actionable information on each athlete's unique strengths and vulnerabilities.

However, sports genomics is not